EDMS 30-312-2

SPECIFICATION

FOR

ROBE OPERATED EXTENDING LADDER

(ALUMINUM / FIBER)

Issue: Feb.-2026/ Rev- 2

- This revision contains option items that must be selected by EDC before bidding.

CONTENTS

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 30-312-2
Robe Operated Extending Ladder (Aluminum / Fiber)	10-02-2026

1. SCOPE

This specification covers the minimum technical requirements for the design, manufacturing, testing, and supplying of robe operated extending ladder. The required is a portable ladder for working with or around electrical network components and buildings of Egyptian Electricity Distribution Companies (EDCs).

2. APPLICABLE STANDARDS

The equipment/material covered in this specification shall comply with the latest edition/amendment of the standards given in Table (1). Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply. Any deviations from the listed standards or the provisions of this specification shall be clearly set out in the offer.

Table (1)

Standard No.	Description
BS EN 131-1	Ladders – Part 1: Terms, types, functional sizes
BS EN 131-2	Ladders – Part 2: Requirements, testing, marking
BS EN 131-3	Ladders – Part 3: Marking and user instructions
BS EN 50528	Insulating ladders for use on or near low voltage electrical installations

3. ENVIRONMENTAL CONDITIONS

The safety behavior of the required ladder shall be guaranteed under the environmental conditions given in Table (2).

Table (2)

Ambient temperature	-5°C to +45°C (50°C is an option according to _ _ EDC requirement)
Maximum relative humidity	95 %
Altitude	Up to 1000 m above mean sea-level
Environment	Humid tropical climate with heavily polluted atmosphere

4. CONSTRUCTION REQUIREMENTS

4.1 Design

- The ladder shall comprise of two parts where upper part shall be extendable by a rope (i.e. two piece rope operated extending ladder).
- Unless otherwise specified by _ _ EDC, the total ladder length shall be 12 m.
- The stile dimensions shall not be less than 80 × 28 mm.
- The distance between two adjacent rungs shall be ranged between 250 – 300 mm.
- The ladder shall be able to bear a maximum total load of 1471 N (150 kg).
- The design shall seek to minimize the existence of shearing and squeeze points-and where they do exist to minimize the shearing and squeezing the effects as far as practicable.
- All connections should be durable and have a strength corresponding to the strain. The connection should be design in a manner that arising notch tensions remain low. The ladder should not bend under heat.
- Screws and nuts should be secured against loosing e.g. by means of self-locking or mechanically locked safety devices.
- Nails are allowed only if their function is related to the production process, e.g. fixation during the drying of glues.
- Welding of joints is permitted if it is as per EN ISO 14731 and EN ISO 3834-1 to EN ISO 3834-4.

4.2 Materials

- According to _ _ EDC requirements, the ladder can be made of Aluminum-alloy (aluminum ladder) or plastics (Fiber ladder).

In case of Aluminum ladder:

- The stiles of ladder shall be made of aluminum-alloy. This alloy shall have an elongation A_5 at rupture measured according to EN ISO 6892-1 of minimum 5%.
- The minimum thickness of all load bearing parts made of aluminum-alloy is 1.5 mm.

In case of Fiber ladder:

- The stiles of ladder shall be made of glass-fiber reinforced plastics. Glass-fiber reinforced plastic shall be protected against penetration of water, dirt, Ultraviolet (UV) radiation, and heat. The surface shall be smooth. The fibers shall be embedded. The Barcol hardness according to EN 59 shall be at least 35.
- The minimum thickness for load-bearing elements is 2 mm.

4.3 Surface Finish

- In order to avoid injuries, accessible edges, corners and protruding parts shall be free of burrs, for example chamfered or rounded.
- Metal parts susceptible to corrosions shall be protected by means of a paint coating or other coating. Under normal conditions aluminum alloys are not susceptible to corrosion. The coating shall be as per BS EN 131-2.
- Wooden parts are not allowed.

4.4 Rungs

- Rungs shall be made of aluminum / aluminum alloy or reinforced UV stabilized plastics and shall have a textured surface on the working face to reduce slipping. The contact surface of the coverings shall adhere firmly to the rungs. Rungs shall be firmly and durably connected to the stiles.
- According to _ _ EDC requirements, rungs can be D-shaped, square or rectangle.
- Rungs shall be replaceable and have a standing surface from front to back of less than 80 mm and at least 20 mm.

- Wooden rungs are not allowed.

4.5 Ladder feet and anti-skid devices

- Bottom ends of ladder shall be slip resistant. A stabilizing bar of suitable width with non-slip rubber feet should be attached to the ladder leg to provide more stability.

4.6 Rung locking devices

- The extension ladders shall be secured from unintentional closing and separation in the position of use.
- All extending ladders shall be fitted with a locking device to keep the ladder hooks engaged on the rung during use.
- Locking devices on rope-operated extending ladders shall reliably ensure a safe catch.
- The rung hooks of rope-operated extension ladders shall be designed in such a way that the upper ladder part cannot fall down by more than one rung if the rope loosens or breaks. This safety requirement shall apply both when the ladder is vertical and in the position of use.
- During use of the ladder the rungs overlapping one another shall be in the same plane perpendicular to the stiles.

4.7 Ropes

- Ropes for extending ladders shall have a minimum strength of 4,000 N. Ropes shall have a minimum diameter of 8 mm. Synthetic ropes shall be stabilized against ultra-violent light.

4.8 Pole grip and lash

- A polyester replaceable belt (35 mm wide, 1.5 m long) fitted with a rapid grip device should be placed above the top rung to secure the ladder to the pole. The part of the ladder in contact with the pole must be covered with electrically insulating rubber.

5. TESTING and INSPECTION

5.1 General

- Ladders shall be subjected to the following tests in accordance with the latest relevant standard and as specified herein.
- All tests shall be carried out by and at the expense of the tenderer who shall supply all the apparatus, instruments and equipment.
- Tests shall be carried out by an accredited independent testing laboratory (or other testing laboratories accepted by _ _ EDC).
- Tests shall be carried out in the presence and under the control of the inspecting engineer of _ _ EDC, except in specified or approved cases where test certificates are accepted. All test results shall be reviewed and approved in writing by _ _ EDC.
- The contractor should advise testing dates. The contractor shall not be entitled to any extension of the time of completion due to the failure of any test or the rejection of any part of the materials as a result of any test.
- Test reports shall clearly show the equipment concerned, the manufacturer's identity, the tests carried out, test results and the standard's requirements against the test results to determine passing or failing of the test.

5.2 Tests

- The following tests shall be carried out for the offered ladders conforming to the latest edition of BS EN 131-2 and BS EN 50528 standards:

  (a) Strength test of stiles

  (b) Bending test of the stiles

  (c) Lateral deflection test of the ladder.

  (d) Bottom stile ends test

  (e) Vertical load on rungs

  (f) Torsion test of rungs

(g) Test for ladder rung hooks

(h) Feet pull test

(i) Maximum extension of ladder

(j) Torsion on ladder length

(k) Rope strength test

(l) Test methods for plastic ladders:

- Thermoset plastics and composite materials
- Reinforced thermoplastics
- Dielectric test

6. SUBMITTALS

6.1 Submittals required with tender:

- One specimen of the offered ladder shall be submitted with the offer for evaluation prior to issuance of the purchase order. The specimen should correspond in all respects with the material they intend to supply. Dimensional verification, indelible markings and finishing shall be checked.
- The supplier shall complete and return one copy of the attached Data Schedule.
- A comprehensive catalogue of the offered ladders. It shall necessarily include details about the ladder size, construction, mechanical properties and material used.
- A valid approval certificate issued by Egyptian Electricity Holding Company.
- Copy of test reports.
- List of deviations & clauses to which exception is taken (if any).

6.2 Submittals required following award of contract:

- Details of manufacturing and testing schedules
- Packing details.
- The supplier should provide all the instructions of use/storing in Arabic.

7. MARKING

– Each ladder shall be clearly, indelibly and permanently marked by the manufacturer and shall include the following basic information:

a) Manufacturer's name.

b) Type of ladder and possible modes of use (description of the type, number and length of the parts, maximum length of ladder in use, maximum standing height measured in position of use according to the recommendation of the manufacturer)

c) Classification of use “professional” or “non-professional” in EN 131-2

d) Standard (include document date) to which the ladder conforms

e) Indication to the necessity for users to read the instructions for use.

f) Month and year of manufacture

g) Weight of the ladder (in kg) and maximal total load (in kg)

h) Insulation if any.

8. GUARANTEE

– The supplier guarantee the supplied materials and all accessories against any defects arising out of faulty design or workmanship, or of defective material for a period of two (2) years from the date of delivery.

– The Supplier should have a dedicated service center in Egypt to cover the product maintenance and provide the technical support.

9. PACKING and DELIVERY

- Ladders shall be packaged in an individual container or package of sufficient strength to properly protect the product from damage. The outside of the container or package shall be marked as specified in relevant standard.
- The type of packaging suitable for transport shall be defined by the manufacturer.
- At the request of the customer or according to government specifications any additional or amended instructions shall be included in the package.

10. TECHNICAL DATA SCHEDULE

- The tenderer must fill in thoroughly the attached technical data schedule.
- Any offer does not accompanied with clear and complete technical data schedule shall be rejected.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 30-312-2
Robe Operated Extending Ladder (Aluminum / Fiber)	10-02-2026

TECHNICAL DATA SCHEDULE

(ALUMINUM / FIBER) LADDER

_ _ EDC Inquiry No:____________________________ Item No:________________

No.	DESCRIPTION	_ _ EDC Requirement*	Vendor Offer**
1	Manufacturer's name		
2	Model		
3	Country of Origin		
4	Applicable Standard	BS EN 131	
5	Maximum total load	150 kg	
6	Length	As per the price schedule	
7	Material of the stile	Glass fiber reinforced plastics / Aluminum alloy	
8	Stile dimensions (L × W × t)	As per clause 4	
9	Material of the rung and the coating	As per clause 4.4	
10	Rung shape	“D” shaped / Square / Rectangle	
11	Rung surface	Anti-slip surface	
12	Distance between two adjacent rungs	250 – 300 mm	
13	Ladder feet	Slip resistant swivel shoe	
14	Locking device provided	Yes, Please refer to clause 4.7	
15	Pulley provided	Yes	
16	**Ladder robe**		
16.1	Diameter	8 mm	
16.2	Minimum strength	4000 N	
17	Insulation between rungs	As per BS EN 131-2	
18	Year and Month/Quarter of Manufacture	Manufactured within two years before the bid closing date.	

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 30-312-2
Robe Operated Extending Ladder (Aluminum / Fiber)	10-02-2026

19	Test Reports	Please refer to clause 5	
20	Submittals	Please refer to clause 6	
21	Marking	Please refer to clause 7	
22	Guarantee	Please refer to clause 8	
23	Packing and delivery	Please refer to clause 9	

(*) - Values to be provided / proposed by the vendor.
(**) – Please, provide explanations for deviations if any. (Use separate sheet if necessary.)

I / We guarantee the technical date given above for the offered material/equipment.

Description	Manufacturer of Material/Equipment	Vendor/Supplier
Vendor/Supplier		
Location and Office Address		
Name and Signature of Authorized Representative with Date		
Official Seal / Stamp		